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0 BEGIN PGM ping MM
1 BLK FORM 0.1 Z X-25 Y-25 Z-30
2 BLK FORM 0.2 X+25 Y+25 Z+0
3 CYCL DEF 247 DATUM SETTING ~
  Q339=+1 ;DATUM NUMBER
4 TOOL CALL 4 Z S1000
5 L Z+200 R0 F2000 M3
6 CYCL DEF 251 RECTANGULAR POCKET ~
  Q215=+0 ;MACHINING OPERATION ~
  Q218=+52 ;FIRST SIDE LENGTH ~
  Q219=+52 ;2ND SIDE LENGTH ~
  Q220=+0 ;CORNER RADIUS ~
  Q368=+0 ;ALLOWANCE FOR SIDE ~
  Q224=+0 ;ANGLE OF ROTATION ~
  Q367=+0 ;POCKET POSITION ~
  Q207=+500 ;FEED RATE FOR MILLNG ~
  Q351=+1 ;CLIMB OR UP-CUT ~
  Q201=-0.3 ;DEPTH ~
  Q202=+5 ;PLUNGING DEPTH ~
  Q369=+0 ;ALLOWANCE FOR FLOOR ~
  Q206=+150 ;FEED RATE FOR PLNGNG ~
  Q338=+0 ;INFEED FOR FINISHING ~
  Q200=+5 ;SET-UP CLEARANCE ~
  Q203=+0 ;SURFACE COORDINATE ~
  Q204=+50 ;2ND SET-UP CLEARANCE ~
  Q370=+1 ;TOOL PATH OVERLAP ~
  Q366=+1 ;PLUNGE ~
  Q385=+500 ;FINISHING FEED RATE
7 CYCL CALL POS X+0 Y+0 Z+0 FMAX
8 CYCL DEF 256 RECTANGULAR STUD ~
  Q218=+48 ;FIRST SIDE LENGTH ~
  Q424=+52 ;WORKPC. BLANK SIDE 1 ~
  Q219=+48 ;2ND SIDE LENGTH ~
  Q425=+52 ;WORKPC. BLANK SIDE 2 ~
  Q220=+0 ;CORNER RADIUS ~
  Q368=+0 ;ALLOWANCE FOR SIDE ~
  Q224=+0 ;ANGLE OF ROTATION ~
  Q367=+0 ;STUD POSITION ~
  Q207=+500 ;FEED RATE FOR MILLNG ~
  Q351=+1 ;CLIMB OR UP-CUT ~
  Q201=-12 ;DEPTH ~
  Q202=+2 ;PLUNGING DEPTH ~
  Q206=+3000 ;FEED RATE FOR PLNGNG ~
  Q200=+2 ;SET-UP CLEARANCE ~
  Q203=+0 ;SURFACE COORDINATE ~
  Q204=+50 ;2ND SET-UP CLEARANCE ~
  Q370=+1 ;TOOL PATH OVERLAP
9 CYCL CALL POS X+0 Y+0 Z+0 FMAX
10 CYCL DEF 251 RECTANGULAR POCKET ~
  Q215=+0 ;MACHINING OPERATION ~
  Q218=+16 ;FIRST SIDE LENGTH ~
  Q219=+16 ;2ND SIDE LENGTH ~
  Q220=+4.5 ;CORNER RADIUS ~
  Q368=+0 ;ALLOWANCE FOR SIDE ~
  Q224=+0 ;ANGLE OF ROTATION ~
  Q367=+0 ;POCKET POSITION ~
  Q207=+500 ;FEED RATE FOR MILLNG ~
  Q351=+1 ;CLIMB OR UP-CUT ~
  Q201=-5 ;DEPTH ~
  Q202=+2 ;PLUNGING DEPTH ~
  Q369=+0 ;ALLOWANCE FOR FLOOR ~
  Q206=+150 ;FEED RATE FOR PLNGNG ~
  Q338=+0 ;INFEED FOR FINISHING ~
  Q200=+2 ;SET-UP CLEARANCE ~
  Q203=+0 ;SURFACE COORDINATE ~
  Q204=+50 ;2ND SET-UP CLEARANCE ~
  Q370=+1 ;TOOL PATH OVERLAP ~
  Q366=+1 ;PLUNGE ~
  Q385=+500 ;FINISHING FEED RATE
11 CYCL CALL POS X+0 Y+0 Z+0 FMAX
12 L Z+200 R0 F2000
13 END PGM ping MM

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0 BEGIN PGM shang MM	Q385=+500 ;FINISHING FEED RATE
1 BLK FORM 0.1 Z X-25 Y-25 Z-30	29 CYCL CALL POS X+0 Y+0 Z+0 FMAX
2 BLK FORM 0.2 X+25 Y+25 Z+0	30 CYCL DEF 253 SLOT MILLING ~
3 CYCL DEF 247 DATUM SETTING ~	Q215=+0 ;MACHINING OPERATION ~
Q339=+1 ;DATUM NUMBER	Q218=+20.5 ;SLOT LENGTH ~
4 TOOL CALL 4 Z S1000	Q219=+8.5 ;SLOT WIDTH ~
5 LBL 1	Q368=+0 ;ALLOWANCE FOR SIDE ~
6 CYCL DEF 7.0 DATUM SHIFT	Q374=+0 ;ANGLE OF ROTATION ~
7 CYCL DEF 7.1 X+0	Q367=+0 ;SLOT POSITION ~
8 CYCL DEF 7.2 Y+0	Q207=+500 ;FEED RATE FOR MILLNG ~
9 CYCL DEF 7.3 Z+0	Q351=+1 ;CLIMB OR UP-CUT ~
10 CYCL DEF 19.0 WORKING PLANE	Q201=-2 ;DEPTH ~
11 CYCL DEF 19.1 A+0 B+0 C+0	Q202=+1 ;PLUNGING DEPTH ~
12 L B+Q121 C+Q122 FMAX	Q369=+0 ;ALLOWANCE FOR FLOOR ~
13 LBL 0	Q206=+150 ;FEED RATE FOR PLNGNG ~
14 L Z+200 R0 F2000 M3	Q338=+0 ;INFEED FOR FINISHING ~
15 LBL 2	Q200=+2 ;SET-UP CLEARANCE ~
16 CYCL DEF 7.0 DATUM SHIFT	Q203=+0 ;SURFACE COORDINATE ~
17 CYCL DEF 7.1 IX+0	Q204=+50 ;2ND SET-UP CLEARANCE ~
18 CYCL DEF 7.2 IY+9	Q366=+1 ;PLUNGE ~
19 CYCL DEF 7.3 IZ+0	Q385=+500 ;FINISHING FEED RATE
20 CYCL DEF 19.0 WORKING PLANE	31 CYCL CALL POS X+0 Y+0 Z+0 FMAX
21 CYCL DEF 19.1 A-15 B+0 C+0	32 L Z+200 R0 F2000
22 CYCL DEF 7.0 DATUM SHIFT	33 CALL LBL 1
23 CYCL DEF 7.1 IX+0	34 END PGM shang MM
24 CYCL DEF 7.2 IY+9	
25 CYCL DEF 7.3 IZ+0	
26 L B+Q121 C+Q122 FMAX	
27 LBL 0	
28 CYCL DEF 251 RECTANGULAR POCKET ~	
Q215=+0 ;MACHINING OPERATION ~	
Q218=+52 ;FIRST SIDE LENGTH ~	
Q219=+30 ;2ND SIDE LENGTH ~	
Q220=+0 ;CORNER RADIUS ~	
Q368=+0 ;ALLOWANCE FOR SIDE ~	
Q224=+0 ;ANGLE OF ROTATION ~	
Q367=+0 ;POCKET POSITION ~	
Q207=+500 ;FEED RATE FOR MILLNG ~	
Q351=+1 ;CLIMB OR UP-CUT ~	
Q201=-1 ;DEPTH ~	
Q202=+5 ;PLUNGING DEPTH ~	
Q369=+0 ;ALLOWANCE FOR FLOOR ~	
Q206=+150 ;FEED RATE FOR PLNGNG ~	
Q338=+0 ;INFEED FOR FINISHING ~	
Q200=+2 ;SET-UP CLEARANCE ~	
Q203=+1 ;SURFACE COORDINATE ~	
Q204=+50 ;2ND SET-UP CLEARANCE ~	
Q370=+1 ;TOOL PATH OVERLAP ~	
Q366=+1 ;PLUNGE ~	

0 BEGIN PGM xia MM	Q385=+500 ;FINISHING FEED RATE
1 BLK FORM 0.1 Z X-25 Y-25 Z-30	29 CYCL CALL POS X+0 Y+0 Z+0 FMAX
2 BLK FORM 0.2 X+25 Y+25 Z+0	30 CYCL DEF 253 SLOT MILLING ~
3 CYCL DEF 247 DATUM SETTING ~	Q215=+0 ;MACHINING OPERATION ~
Q339=+1 ;DATUM NUMBER	Q218=+20.5 ;SLOT LENGTH ~
4 TOOL CALL 4 Z S1000	Q219=+8.5 ;SLOT WIDTH ~
5 LBL 1	Q368=+0 ;ALLOWANCE FOR SIDE ~
6 CYCL DEF 7.0 DATUM SHIFT	Q374=+0 ;ANGLE OF ROTATION ~
7 CYCL DEF 7.1 X+0	Q367=+0 ;SLOT POSITION ~
8 CYCL DEF 7.2 Y+0	Q207=+500 ;FEED RATE FOR MILLNG ~
9 CYCL DEF 7.3 Z+0	Q351=+1 ;CLIMB OR UP-CUT ~
10 CYCL DEF 19.0 WORKING PLANE	Q201=-2 ;DEPTH ~
11 CYCL DEF 19.1 A+0 B+0 C+0	Q202=+1 ;PLUNGING DEPTH ~
12 L B+Q121 C+Q122 FMAX	Q369=+0 ;ALLOWANCE FOR FLOOR ~
13 LBL 0	Q206=+150 ;FEED RATE FOR PLNGNG ~
14 L Z+200 R0 F2000 M3	Q338=+0 ;INFEED FOR FINISHING ~
15 LBL 2	Q200=+2 ;SET-UP CLEARANCE ~
16 CYCL DEF 7.0 DATUM SHIFT	Q203=+0 ;SURFACE COORDINATE ~
17 CYCL DEF 7.1 IX+0	Q204=+50 ;2ND SET-UP CLEARANCE ~
18 CYCL DEF 7.2 IY-9	Q366=+1 ;PLUNGE ~
19 CYCL DEF 7.3 IZ+0	Q385=+500 ;FINISHING FEED RATE
20 CYCL DEF 19.0 WORKING PLANE	31 CYCL CALL POS X+0 Y+0 Z+0 FMAX
21 CYCL DEF 19.1 A+15 B+0 C+0	32 L Z+200 R0 F2000
22 CYCL DEF 7.0 DATUM SHIFT	33 CALL LBL 1
23 CYCL DEF 7.1 IX+0	34 END PGM xia MM
24 CYCL DEF 7.2 IY-9	
25 CYCL DEF 7.3 IZ+0	
26 L B+Q121 C+Q122 FMAX	
27 LBL 0	
28 CYCL DEF 251 RECTANGULAR POCKET ~	
Q215=+0 ;MACHINING OPERATION ~	
Q218=+52 ;FIRST SIDE LENGTH ~	
Q219=+30 ;2ND SIDE LENGTH ~	
Q220=+0 ;CORNER RADIUS ~	
Q368=+0 ;ALLOWANCE FOR SIDE ~	
Q224=+0 ;ANGLE OF ROTATION ~	
Q367=+0 ;POCKET POSITION ~	
Q207=+500 ;FEED RATE FOR MILLNG ~	
Q351=+1 ;CLIMB OR UP-CUT ~	
Q201=-1 ;DEPTH ~	
Q202=+5 ;PLUNGING DEPTH ~	
Q369=+0 ;ALLOWANCE FOR FLOOR ~	
Q206=+150 ;FEED RATE FOR PLNGNG ~	
Q338=+0 ;INFEED FOR FINISHING ~	
Q200=+2 ;SET-UP CLEARANCE ~	
Q203=+1 ;SURFACE COORDINATE ~	
Q204=+50 ;2ND SET-UP CLEARANCE ~	
Q370=+1 ;TOOL PATH OVERLAP ~	
Q366=+1 ;PLUNGE ~	

0 BEGIN PGM zuo MM	Q385=+500 ;FINISHING FEED RATE
1 BLK FORM 0.1 Z X-25 Y-25 Z-30	29 CYCL CALL POS X+0 Y+0 Z+0 FMAX
2 BLK FORM 0.2 X+25 Y+25 Z+0	30 CYCL DEF 257 CIRCULAR STUD ~
3 CYCL DEF 247 DATUM SETTING ~ Q339=+1 ;DATUM NUMBER	Q223=+10 ;FINISHED PART DIA. ~
4 TOOL CALL 5 Z S1000	Q222=+52 ;WORKPIECE BLANK DIA. ~
5 LBL 1	Q368=+0 ;ALLOWANCE FOR SIDE ~
6 CYCL DEF 7.0 DATUM SHIFT	Q207=+500 ;FEED RATE FOR MILLNG ~
7 CYCL DEF 7.1 X+0	Q351=+1 ;CLIMB OR UP-CUT ~
8 CYCL DEF 7.2 Y+0	Q201=-1 ;DEPTH ~
9 CYCL DEF 7.3 Z+0	Q202=+5 ;PLUNGING DEPTH ~
10 CYCL DEF 19.0 WORKING PLANE	Q206=+3000 ;FEED RATE FOR PLNGNG ~
11 CYCL DEF 19.1 A+0 B+0 C+0	Q200=+2 ;SET-UP CLEARANCE ~
12 L B+Q121 C+Q122 FMAX	Q203=+1 ;SURFACE COORDINATE ~
13 LBL 0	Q204=+50 ;2ND SET-UP CLEARANCE ~
14 L Z+200 R0 F2000 M3	Q370=+1 ;TOOL PATH OVERLAP
15 LBL 2	31 CYCL CALL POS X+0 Y+0 Z+0 FMAX
16 CYCL DEF 7.0 DATUM SHIFT	32 L Z+200 R0 F2000
17 CYCL DEF 7.1 IX-9	33 CALL LBL 1
18 CYCL DEF 7.2 IY+0	34 TOOL CALL 3 Z S800
19 CYCL DEF 7.3 IZ+0	35 CALL LBL 1
20 CYCL DEF 19.0 WORKING PLANE	36 L Z+200 R0 F2000 M3
21 CYCL DEF 19.1 A+0 B-15 C+0	37 CALL LBL 2
22 CYCL DEF 7.0 DATUM SHIFT	38 CYCL DEF 200 DRILLING ~
23 CYCL DEF 7.1 IX-9	Q200=+5 ;SET-UP CLEARANCE ~
24 CYCL DEF 7.2 IY+0	Q201=-8 ;DEPTH ~
25 CYCL DEF 7.3 IZ+0	Q206=+150 ;FEED RATE FOR PLNGNG ~
26 L B+Q121 C+Q122 FMAX	Q202=+2 ;PLUNGING DEPTH ~
27 LBL 0	Q210=+0 ;DWELL TIME AT TOP ~
28 CYCL DEF 251 RECTANGULAR POCKET ~	Q203=+0 ;SURFACE COORDINATE ~
Q215=+0 ;MACHINING OPERATION ~	Q204=+50 ;2ND SET-UP CLEARANCE ~
Q218=+30 ;FIRST SIDE LENGTH ~	Q211=+0 ;DWELL TIME AT DEPTH
Q219=+52 ;2ND SIDE LENGTH ~	39 CYCL CALL POS X+0 Y+0 Z+0 FMAX
Q220=+0 ;CORNER RADIUS ~	40 L Z+200 R0 F2000
Q368=+0 ;ALLOWANCE FOR SIDE ~	41 CALL LBL 1
Q224=+0 ;ANGLE OF ROTATION ~	42 END PGM zuo MM
Q367=+0 ;POCKET POSITION ~	
Q207=+500 ;FEED RATE FOR MILLNG ~	
Q351=+1 ;CLIMB OR UP-CUT ~	
Q201=-1 ;DEPTH ~	
Q202=+5 ;PLUNGING DEPTH ~	
Q369=+0 ;ALLOWANCE FOR FLOOR ~	
Q206=+150 ;FEED RATE FOR PLNGNG ~	
Q338=+0 ;INFEEED FOR FINISHING ~	
Q200=+2 ;SET-UP CLEARANCE ~	
Q203=+2 ;SURFACE COORDINATE ~	
Q204=+50 ;2ND SET-UP CLEARANCE ~	
Q370=+1 ;TOOL PATH OVERLAP ~	
Q366=+1 ;PLUNGE ~	

0 BEGIN PGM you MM	Q385=+500 ;FINISHING FEED RATE
1 BLK FORM 0.1 Z X-25 Y-25 Z-30	29 CYCL CALL POS X+0 Y+0 Z+0 FMAX
2 BLK FORM 0.2 X+25 Y+25 Z+0	30 CYCL DEF 257 CIRCULAR STUD ~
3 CYCL DEF 247 DATUM SETTING ~	Q223=+10 ;FINISHED PART DIA. ~
Q339=+1 ;DATUM NUMBER	Q222=+52 ;WORKPIECE BLANK DIA. ~
4 TOOL CALL 5 Z S1000	Q368=+0 ;ALLOWANCE FOR SIDE ~
5 LBL 1	Q207=+500 ;FEED RATE FOR MILLNG ~
6 CYCL DEF 7.0 DATUM SHIFT	Q351=+1 ;CLIMB OR UP-CUT ~
7 CYCL DEF 7.1 X+0	Q201=-1 ;DEPTH ~
8 CYCL DEF 7.2 Y+0	Q202=+5 ;PLUNGING DEPTH ~
9 CYCL DEF 7.3 Z+0	Q206=+3000 ;FEED RATE FOR PLNGNG ~
10 CYCL DEF 19.0 WORKING PLANE	Q200=+2 ;SET-UP CLEARANCE ~
11 CYCL DEF 19.1 A+0 B+0 C+0	Q203=+1 ;SURFACE COORDINATE ~
12 L B+Q121 C+Q122 FMAX	Q204=+50 ;2ND SET-UP CLEARANCE ~
13 LBL 0	Q370=+1 ;TOOL PATH OVERLAP
14 L Z+200 R0 F2000 M3	31 CYCL CALL POS X+0 Y+0 Z+0 FMAX
15 LBL 2	32 L Z+200 R0 F2000
16 CYCL DEF 7.0 DATUM SHIFT	33 CALL LBL 1
17 CYCL DEF 7.1 IX+9	34 TOOL CALL 3 Z S800
18 CYCL DEF 7.2 IY+0	35 CALL LBL 1
19 CYCL DEF 7.3 IZ+0	36 L Z+200 R0 F2000 M3
20 CYCL DEF 19.0 WORKING PLANE	37 CALL LBL 2
21 CYCL DEF 19.1 A+0 B+15 C+0	38 CYCL DEF 200 DRILLING ~
22 CYCL DEF 7.0 DATUM SHIFT	Q200=+5 ;SET-UP CLEARANCE ~
23 CYCL DEF 7.1 IX+9	Q201=-8 ;DEPTH ~
24 CYCL DEF 7.2 IY+0	Q206=+150 ;FEED RATE FOR PLNGNG ~
25 CYCL DEF 7.3 IZ+0	Q202=+2 ;PLUNGING DEPTH ~
26 L B+Q121 C+Q122 FMAX	Q210=+0 ;DWELL TIME AT TOP ~
27 LBL 0	Q203=+0 ;SURFACE COORDINATE ~
28 CYCL DEF 251 RECTANGULAR POCKET ~	Q204=+50 ;2ND SET-UP CLEARANCE ~
Q215=+0 ;MACHINING OPERATION ~	Q211=+0 ;DWELL TIME AT DEPTH
Q218=+30 ;FIRST SIDE LENGTH ~	39 CYCL CALL POS X+0 Y+0 Z+0 FMAX
Q219=+52 ;2ND SIDE LENGTH ~	40 L Z+200 R0 F2000
Q220=+0 ;CORNER RADIUS ~	41 CALL LBL 1
Q368=+0 ;ALLOWANCE FOR SIDE ~	42 END PGM you MM
Q224=+0 ;ANGLE OF ROTATION ~	
Q367=+0 ;POCKET POSITION ~	
Q207=+500 ;FEED RATE FOR MILLNG ~	
Q351=+1 ;CLIMB OR UP-CUT ~	
Q201=-1 ;DEPTH ~	
Q202=+5 ;PLUNGING DEPTH ~	
Q369=+0 ;ALLOWANCE FOR FLOOR ~	
Q206=+150 ;FEED RATE FOR PLNGNG ~	
Q338=+0 ;INFEED FOR FINISHING ~	
Q200=+2 ;SET-UP CLEARANCE ~	
Q203=+2 ;SURFACE COORDINATE ~	
Q204=+50 ;2ND SET-UP CLEARANCE ~	
Q370=+1 ;TOOL PATH OVERLAP ~	
Q366=+1 ;PLUNGE ~	

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0 BEGIN PGM zong MM
1 CALL PGM TNC:\xl4\ping.h
2 CALL PGM TNC:\xl4\shang.h
3 CALL PGM TNC:\xl4\xia.h
4 CALL PGM TNC:\xl4\zuo.h
5 CALL PGM TNC:\xl4\you.h
6 END PGM zong MM
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